motor, like a camera's tripod. Attach one end of one of the metallic rods to the shaft of the second servo motor and the remaining two rods to the head of the robot, as shown in Figs 5 and 6.

(***Note.*** In the shown project, servo motors were used as support; jaw and head movement was not implemented.)

The final robot head with eyes using two OLED display modules will look like the one in Fig. 7.

Coding

Coding has two parts: Coding for the robot's eyes using Arduino and coding for face recognition using Raspberry Pi.

Coding for robot's eyes. Before beginning with the Arduino code (smartface_recog.ino), go to the Library Manager of Arduino IDE and install the following libraries:

- Adafruit GFX
- SSD1306 OLED

Add the above Arduino libraries into the code using the 'include' function and then insert the bitmap hexadecimal code for the eyes, as shown in Fig. 8.

Next, create a loop function to call the bitmap codes to preview these on the OLED display. To make the eyes more interactive and lively, use a random () function generator (refer Fig. 9).

Coding for face recognition. Raspberry Pi is used to recognise the person in front of the robot (known or unknown). Install the following libraries in Raspberry Pi for the Python3 environment:

- Numpy
- Scipy
- Open cv
- Cv2
- Face recognition
- Espeak
- Matplot
- Keras

To install these libraries, follow the library installation instructions available in the documentation folder of each library.

Import three modules in the Python code: face recognition, cv2,

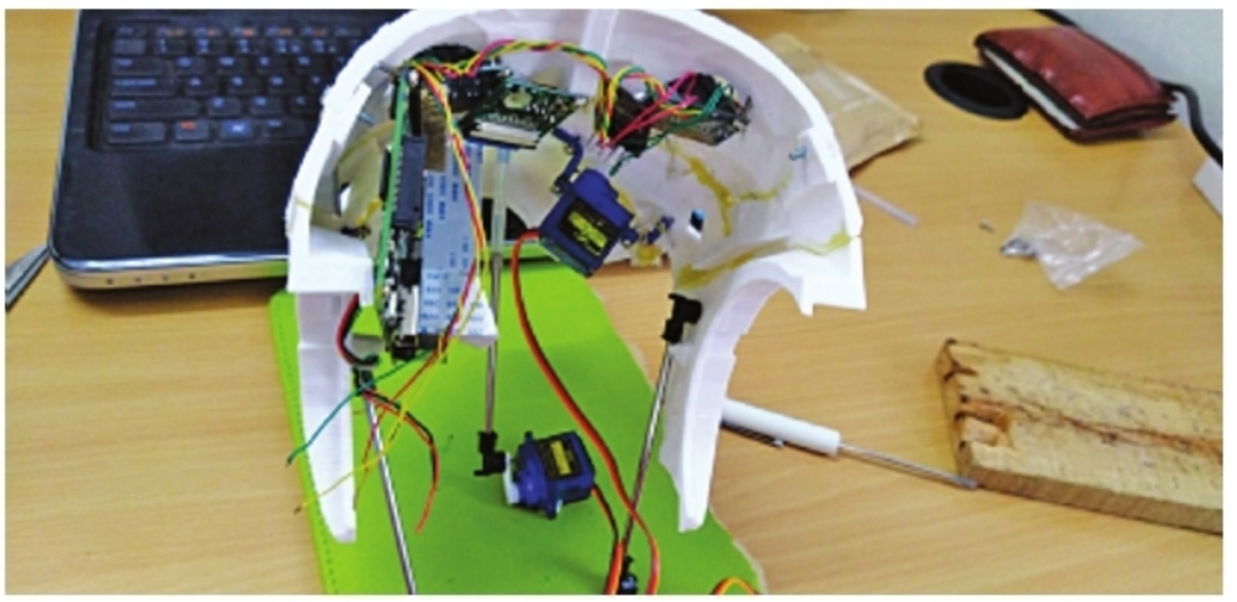


Fig. 6: Three metallic rods for support

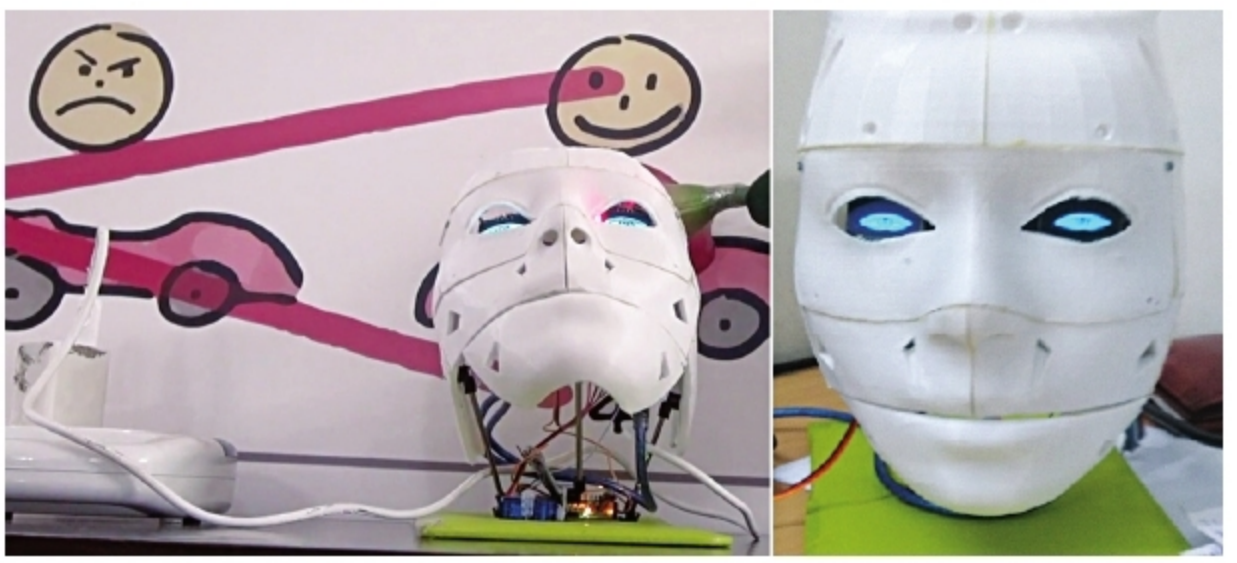


Fig. 7: Robot head with OLED eyes

```
 long randNumber;

#include "ssd1306.h"
#include "Adafruit_GFX.h"
const uint8_t Owl [] PROGMEM = {
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x40. 0x00. 0x00. 0x20. 0x20. 0x80. 0x80. 0x80. 0x80. 0x80.
```

Fig. 8: Bitmap hexadecimal in Arduino code

```
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 robrt

}


void loop()
{
   randNumber = random(100, 2000);
   ssd1306_drawBitmap(0, 0, 128, 64, Owl);
  delay( randNumber);
    ssd1306_drawBitmap(0, 0, 128, 64, Ow);
     delay( randNumber);
      ssd1306_drawBitmap(0, 0, 128, 64, Ow2);
     delay(randNumber);

       ssd1306_drawBitmap(0, 0, 128, 64, Ow4);
  delay( randNumber);
    ssd1306_drawBitmap(0, 0, 128, 64, Ow5);
     delay(randNumber);
      ssd1306_drawBitmap(0, 0, 128, 64, Ow6);
     delay(randNumber);
      ssd1306_drawBitmap(0, 0, 128, 64, Ow7);
     delay(randNumber);
      ssd1306_drawBitmap(0, 0, 128, 64, Ow8);
  delay( randNumber);
    ssd1306_drawBitmap(0, 0, 128, 64, Ow9);
     delay(randNumber);
      ssd1306_drawBitmap(0, 0, 128, 64, Ow10);
     delay(randNumber);
```

Fig. 9: Random function generator